

MEASUREMENTS OF TRANSVERSE SPIN  
DEPENDENT  $\pi^+\pi^-$  AZIMUTHAL  
CORRELATION ASYMMETRY AND  
UNPOLARIZED  $\pi^+\pi^-$  CROSS SECTION IN  $pp$   
COLLISIONS AT  $\sqrt{s} = 200$  GeV AT STAR

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A Dissertation  
Submitted to  
the Temple University Graduate Board

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In Partial Fulfillment  
of the Requirements for the Degree  
DOCTOR OF PHILOSOPHY

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by  
Babu R. Pokhrel  
December 9, 2022

Examining Committee Members:

Bernd Surrow, Department of Physics, Advisory Chair  
Andreas Metz, Department of Physics  
Jim Napolitano, Department of Physics  
TBD, External Reader

# ABSTRACT

The transversity distribution function,  $h_1^q(x)$ , where  $x$  is the longitudinal momentum fraction of the proton carried by quark  $q$ , encodes the proton's transverse spin structure at the leading twist. Extraction of it is difficult because of its chiral-odd nature. However, it can be coupled with a spin-dependent interference fragmentation function (FF),  $H_1^{\triangleleft,q}$ , in polarized proton-proton ( $p^\uparrow p$ ) collisions. The coupling of  $h_1^q(x)$  and  $H_1^{\triangleleft,q}$  produces an experimentally measurable azimuthal correlation asymmetry,  $A_{UT}$ , between the spin of the fragmenting quark and the final state di-hadron. A model-independent extraction of transversity from these measurements relies on the knowledge of di-hadron FFs, namely the unpolarized di-hadron FFs,  $D_1^{h_1 h_2/q(g)}$  for quarks,  $q$ (gluons,  $g$ ). Extraction of these FFs requires measurements of the unpolarized di-hadron cross section in  $pp$  collisions, which are desperately needed. In  $pp$  collisions, the cross section measurement provides access to the  $D_1^{h_1 h_2}$  for both, the quarks and gluons. This thesis outlines the measurements of di-pion azimuthal correlation asymmetry in the mid-pseudorapidity region using the RHIC run 2015 polarized  $pp$  data at  $\sqrt{s} = 200$  GeV and the measurement of the unpolarized di-pion cross section measurement in the invariant mass bins in forward and backward pseudorapidity region with respect to the polarized beam using the RHIC run 2012  $pp$  data at  $\sqrt{s} = 200$  GeV.

# ACKNOWLEDGEMENTS

I found very hard to write something in this part of my whole thesis, because there were many people around me who has huge direct and indirect influence over my PhD career and beyond. I would like to thank my advisor Dr. Bernd Surrow on believing and guiding throughout this journey. I am thankful to Dr. Andreas Metz and Dr. Jim Napolitano, who read my thesis as the thesis committee. Moreover, I had an opportunity to take Classical Mechanics and Particle Physics courses in my first and forth semester with Dr. Metz and first two semester of quantum mechanics courses with Dr. Napolitano. I am thankful to the whole physics department and administration for all the administrative help through one of the toughest journey of my life.

I would like to thank NUPAX group, particle physics research group lead by Dr. Surrow. Being a part of it was already a happy starting of the career as the group is packed with the hardworking particle physics enthusiast and experts. Continuous discussion within the group helped resolving many research obstacles that lead to the fruitful outcomes. It's hard to forget to thank Dr. Matt Posik, Dr. Jae Nam, Dr. Amilkar Quintaro (former NUPAX member), and Navagyan Ghimire, you have motivated me a lot!

My parents are "idol" for my academic career: they didn't compromise on giving good education to their childrens besides the financial hardship. They knew that how hard the life is to be illiterate. I am very fortunate and lucky to have you as my parents. My wife was always there by my side motivating and feeling herself proud of me. I am blessed to have such a caring and loving wife, without her I could imagine how rough my life would be.

Cheers to all my friends, you all should be proud of me!

# TABLE OF CONTENTS

|                                                                   |           |
|-------------------------------------------------------------------|-----------|
| <b>ABSTRACT</b>                                                   | <b>i</b>  |
| <b>ACKNOWLEDGEMENTS</b>                                           | <b>ii</b> |
| <b>LIST OF FIGURES</b>                                            | <b>v</b>  |
| <b>LIST OF TABLES</b>                                             | <b>vi</b> |
| <br><b>CHAPTER</b>                                                |           |
| <b>1 INTRODUCTION</b>                                             | <b>1</b>  |
| 1.1 Elementary Particles And Standard Model . . . . .             | 3         |
| <b>2 THEORITICAL BACKGROUND</b>                                   | <b>5</b>  |
| 2.1 Elementary particles and Standard model . . . . .             | 5         |
| 2.2 Parton distribution functions . . . . .                       | 5         |
| 2.3 Fragmentation Functions . . . . .                             | 5         |
| <b>3 Observables for transversity PDF</b>                         | <b>6</b>  |
| 3.1 Dihadron correlation asymmetry . . . . .                      | 6         |
| 3.2 dihadron cross-section . . . . .                              | 6         |
| <b>4 Experiment and dataset</b>                                   | <b>7</b>  |
| 4.1 Run 15 pp dataset for the asymmetry measurement . . . . .     | 7         |
| 4.2 Run 12 pp dataset for the cross section measurement . . . . . | 7         |
| 4.3 Event selection and di-pion construction . . . . .            | 7         |

|          |                                  |           |
|----------|----------------------------------|-----------|
| <b>5</b> | <b>Analysis</b>                  | <b>8</b>  |
| 5.1      | Asymmetry measurement . . . . .  | 8         |
| 5.2      | Dihadron cross section . . . . . | 8         |
| <b>6</b> | <b>Conclusion and outlook</b>    | <b>9</b>  |
|          | <b>BIBLIOGRAPHY</b>              | <b>10</b> |
|          | <b>APPENDICES</b>                |           |
| <b>A</b> | <b>SOME ADDITIONAL STUFF</b>     | <b>11</b> |
| <b>B</b> | <b>MORE ADDITIONAL STUFF</b>     | <b>12</b> |

# LIST OF FIGURES

|     |                                                           |   |
|-----|-----------------------------------------------------------|---|
| 1.1 | Visualization of elementary particle's dimension. . . . . | 2 |
| 1.2 | Fundamental particles. . . . .                            | 3 |
| 1.3 | Forces in nature. . . . .                                 | 4 |

# LIST OF TABLES

# CHAPTER 1

## INTRODUCTION

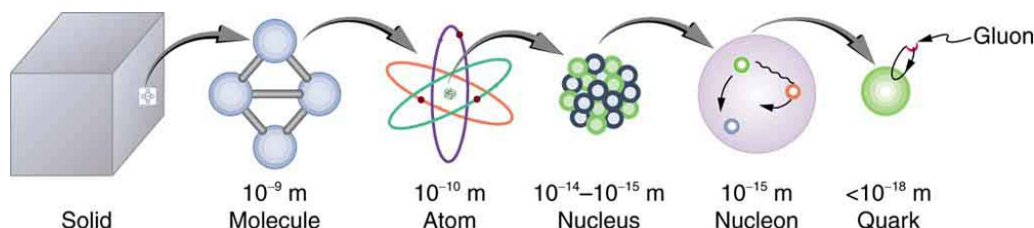
It was fascinating to know that the matters surrounding us are composed of many smaller components, which we cannot really see by our naked eyes. At elementary school, we learned about the substructure of matters, an elements, molecules, and atoms. At high school, it went further deeper to the structure of an atom and its sustructure: an electon, proton, and neutron. Initially, the electron, proton, and neutron were thought to be the fundamental particles, an unbreakable without substructures, until the discovery that confirms that the proron is not a fundamental: it has something more fundamental structure within. This leads to the discovery of the quarks and gluons: fundamental particles which so far never observed as free particle in nature. The development of particles discovery motivates physicists to work on the behavior of these which should be able to explain the property of matter that evolves as a composite effect of the fundamental particle constituents. The theory of quantum electrodynamics (QED) and quantum chromodynamics (QCD) are of huge success so far, not complete though, which explains the bound structure to form nucleons and their interactions within nucleons. Before talking about the fundamental theories of particle physics, we should be aware of the size of nucleons and elementary particles that we are talking about and how particle physicists are actually “see” these particles.

The substructure a matter can be decomposed in to its simplest unbreakable limit, the elementary particles, as shown in figure 1.1. If we are talking of quarks, the fundamental present in nucleons<sup>1</sup>, we are talking about the extremely small dimension of

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<sup>1</sup>protons and neutrons

$\sim 10^{-18}\text{m}$ . So, to understand the property of matter one should understand the nature and dynamics of quarks and gluons which form a bound state nucleons. To open up a nucleon to study its constituents properties is not as simple as that. However, particle physicists have come up with the idea of particle collider and accelerators, which are capable of producing and smashing particles, of enough energy, to open it up. By analyzing debris coming out of collision, physicist can actually unravel the inner structure. The energy of particles is crucial in particle collider, as we know that in order to probe nucleon (of size  $\sim 10^{-18}\text{ m}$ ), the wavelength of the probe should be less than the de-broglie wavelength<sup>2</sup>. During collision, particles talk to each other, we called it particle interaction, exchanging intermediate bosons. If the energy is sufficient enough, the imparted parton knocked off the nucleon and form a debris, more stable final state particles, which will be detected by the particle detectors.



**Figure 1.1:** Visualization of elementary particle's dimension.

Generally, higher the energy of the smashing particles, the higher mass particles are produced. The lighter mass particles were discovered followed by the heavier mass particles; the more advanced and high energy particle collider are in use in modern days. The particles produced due to collision are detected by the particle detectors, which usually measures particle charge, momentum, identify particles. Studying those particles, the internal structure of nucleon, such as momentum, spin distribution of quarks, can be probed, which forms the bound state of nucleons.

The outline of thesis is as follows: in the first section I will discussed about elementary particles, their properties, and the corresponding theory that describes them: the standard model (SM). The structure of proton, described in terms of parton distribution functions (PDFs) will be discussed in section ???. The STAR experiment and the dataset will be discussed in section ??. In the section ?? and ??, the detailed analysis of the dihadron correlation asymmetries and the cross section

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<sup>2</sup> $\lambda = h/p$ , where  $h$  = planck's constant,  $p$  = momentum

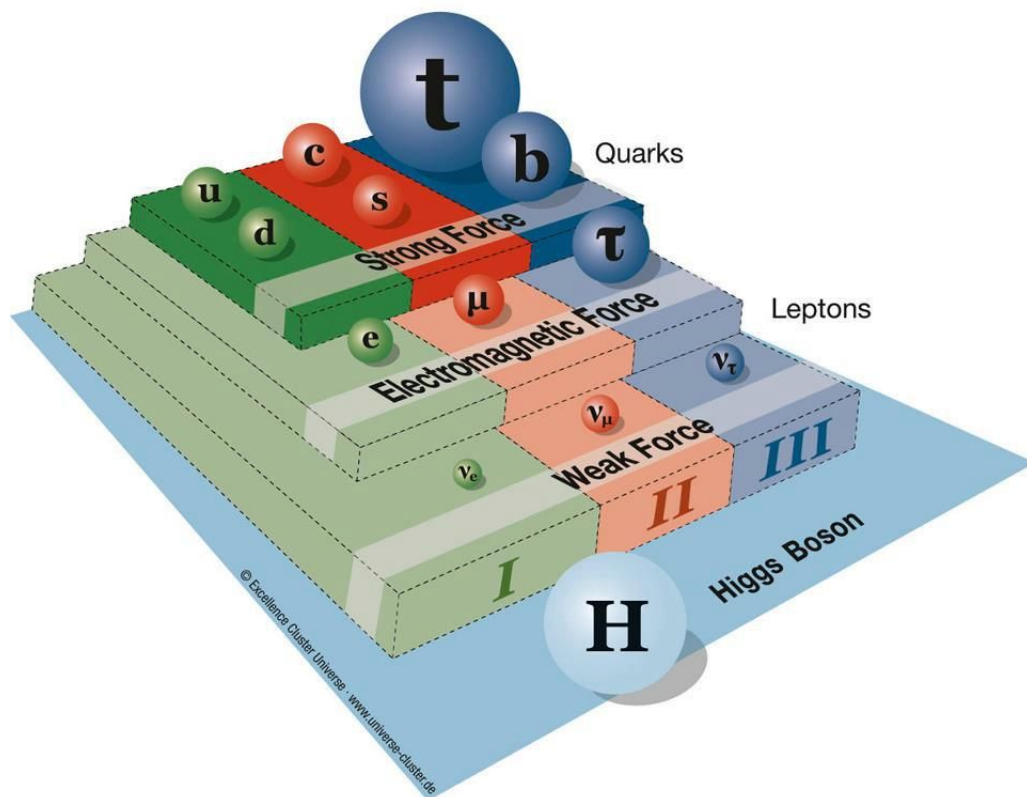
will be discussed. Finally, with the Summary and Outlook chapter, this thesis will be concluded.

## 1.1 Elementary Particles And Standard Model

The standard model describes all the elementary particles and the forces that binds those to form a bound state, which eventually comprises all the visible matters in the world. Based on the spin, particles are classified in to fermions and bosons. Fermions are spin- $\frac{1}{2}$  particles, whereas bosons are particles with integer spin. Bosons comes in between fermions to hold them together as a glue to form a composite structure of quarks. Depending on the quark content, those stable particles are further clasified in to mesons and baryons. Mesons are of di-quark state ( $q\bar{q}$ ) whereas baryons are tri-quark state (combination of quarks and/or antiquarks). Question may arise that doesn't bosons violate Pauli's exclusion priciples, since identicle particles cannot occupy the same quantum state? Here comes the concept of the color charge as a saviour.

| Three generations of matter (fermions) |                                                                                                |                                                                                                |                                                                                                |                                                                                 |                         |
|----------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------|
|                                        | I                                                                                              | II                                                                                             | III                                                                                            |                                                                                 |                         |
| mass→                                  | 2.4 MeV/c <sup>2</sup>                                                                         | 1.27 GeV/c <sup>2</sup>                                                                        | 171.2 GeV/c <sup>2</sup>                                                                       | 0                                                                               | ? GeV/c <sup>2</sup>    |
| charge→                                | $\frac{2}{3}$                                                                                  | $\frac{2}{3}$                                                                                  | $\frac{2}{3}$                                                                                  | 0                                                                               | 0                       |
| spin→                                  | $\frac{1}{2}$                                                                                  | $\frac{1}{2}$                                                                                  | $\frac{1}{2}$                                                                                  | 1                                                                               | 0                       |
| name→                                  | <b>u</b><br>up                                                                                 | <b>c</b><br>charm                                                                              | <b>t</b><br>top                                                                                | <b><math>\gamma</math></b><br>photon                                            | <b>H</b><br>Higgs boson |
| Quarks                                 | 4.8 MeV/c <sup>2</sup><br>$-\frac{1}{3}$<br>$\frac{1}{2}$<br><b>d</b><br>down                  | 104 MeV/c <sup>2</sup><br>$-\frac{1}{3}$<br>$\frac{1}{2}$<br><b>s</b><br>strange               | 4.2 GeV/c <sup>2</sup><br>$-\frac{1}{3}$<br>$\frac{1}{2}$<br><b>b</b><br>bottom                | 0<br>0<br>1<br><b>g</b><br>gluon                                                |                         |
|                                        | <2.2 eV/c <sup>2</sup><br>0<br>$\frac{1}{2}$<br><b><math>\nu_e</math></b><br>electron neutrino | <0.17 MeV/c <sup>2</sup><br>0<br>$\frac{1}{2}$<br><b><math>\nu_\mu</math></b><br>muon neutrino | <15.5 MeV/c <sup>2</sup><br>0<br>$\frac{1}{2}$<br><b><math>\nu_\tau</math></b><br>tau neutrino | 91.2 GeV/c <sup>2</sup><br>0<br>1<br><b><math>Z^0</math></b><br>Z boson         |                         |
|                                        | 0.511 MeV/c <sup>2</sup><br>-1<br>$\frac{1}{2}$<br><b>e</b><br>electron                        | 105.7 MeV/c <sup>2</sup><br>-1<br>$\frac{1}{2}$<br><b><math>\mu</math></b><br>muon             | 1.777 GeV/c <sup>2</sup><br>-1<br>$\frac{1}{2}$<br><b><math>\tau</math></b><br>tau             | 80.4 GeV/c <sup>2</sup><br>$\pm 1$<br>1<br><b><math>W^\pm</math></b><br>W boson |                         |
| Leptons                                |                                                                                                |                                                                                                |                                                                                                | Gauge bosons                                                                    |                         |

Figure 1.2: Fundamental particles.



**Figure 1.3:** Forces in nature.

# CHAPTER 2

## THEORETICAL BACKGROUND

### 2.1 Elementary particles and Standard model

Elementary particles classification, QED, QCD, electroweak process including picture of standard model particle classification.

### 2.2 Parton distribution functions

Describe all PDFs and their current experimental and global fit status with their significance.

### 2.3 Fragmentation Functions

Discuss all the fragmentations and their significance ..

# CHAPTER 3

## Observables for transversity PDF

Discuss observables that are required for the transversity through other channel, for example, Collins effect, Sivers's effect, drell-yan channel,  $e^+e^-$  channel, etc. And discuss extensively the dihadron channel for the asymmetry and cross section.

### 3.1 Dihadron correlation asymmetry

Detailed derivation of the cross ratio method and its elegance.

### 3.2 dihadron cross-section

dihadron cross section measurement: why we need this observable? How we are getting this so far?

# CHAPTER 4

## Experiment and dataset

Explain in details about the RHIC, STAR detectors and mention about the space charge calibration, the details of which will be included in the appendix.

**4.1 Run 15 pp dataset for the asymmetry measurement**

**4.2 Run 12 pp dataset for the cross section measurement**

**4.3 Event selection and di-pion construction**

# CHAPTER 5

## Analysis

5.1 Asymmetry measurement

5.2 Dihadron cross section

# CHAPTER 6

## Conclusion and outlook

# BIBLIOGRAPHY

# **APPENDIX A**

## **SOME ADDITIONAL STUFF**

Appendices go at the end.

# **APPENDIX B**

## **MORE ADDITIONAL STUFF**

This is another appendix.